

ACE: Agentic Control for Embodied Manipulation via Zero-shot Workflow Reasoning

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Abstract: Open-ended tabletop manipulation requires agents to not only understand natural language but also adapt to dynamic environments and execution failures. We present **ACE** (**A**gentic **C**ontrol for **E**mbodied manipulation), a zero-shot workflow reasoning framework for tabletop pick-and-place from natural language. Rather than relying on direct low-level action mapping, ACE combines agentic workflow reasoning with two robot-facing executable skills: a visual grounding interface and a reusable pick-and-place primitive. To bridge semantic reasoning and physical control, the active sub-goal is grounded into a **mask-mediated vision-action interface**. This unified mask specifies the target object and destination, is tracked over time, exposed for human verification, and ultimately passed to a task-agnostic downstream policy for execution. Crucially, ACE operates in a closed loop supported by a **multi-timescale memory**. After an action is executed, the system automatically verifies whether the intended sub-goal succeeded, using the outcome to advance, retry, repair, or replan. This enables online adaptation to user corrections, scene changes, and physical failures. We evaluate ACE on logically complex, long-horizon tasks, including zero-shot multi-step equation formation with number cubes and constraint-based object retrieval. ACE demonstrates task-level zero-shot generalization on novel semantic constraints and randomized tabletop scenes without task-specific retraining. Specifically, while standard end-to-end baselines struggle to complete these logically demanding tasks, ACE achieves a 50% success rate in equation formation and a 70% success rate in constraint retrieval. This contrast demonstrates that explicit workflow reasoning and mask-mediated control offer a robust, practical route toward adaptable robotic manipulation.

Keywords: Embodied reasoning, Robot manipulation, Zero-shot planning

1 Introduction

Natural language provides an intuitive interface for open-ended robotic control, allowing users to specify high-level goals and refine intent dynamically. However, this flexibility makes execution inherently difficult. Instructions are often underspecified, and task constraints may emerge only after execution begins [1, 2, 3]. This challenge is particularly visible in tabletop pick-and-place. While a basic robotic primitive, executing open-ended tasks—such as forming equations from number cubes based on abstract constraints—requires more than recognizing objects and executing grasps. Traditional manipulation systems often rely on task-specific policies or libraries of fixed rules [4, 5], which struggle to generalize when goals vary significantly across episodes. To succeed, a system

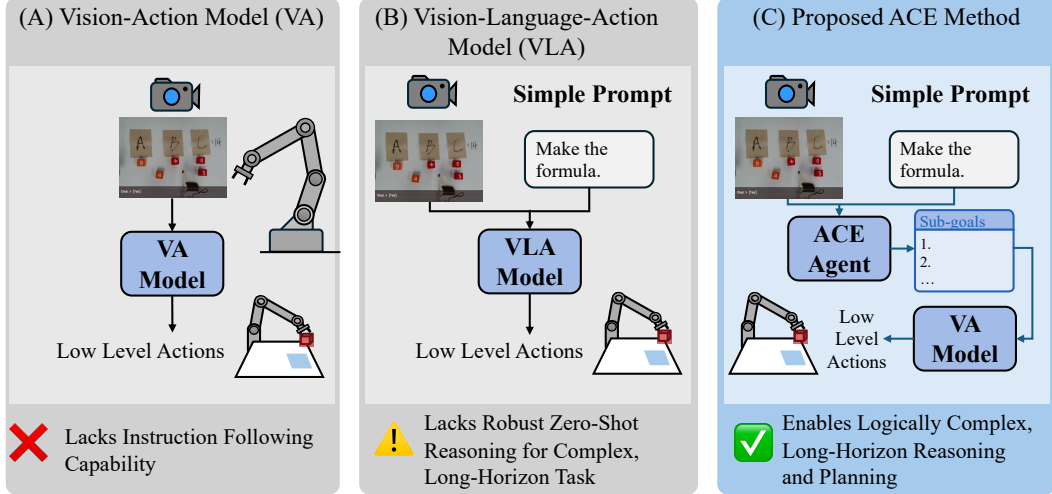


Figure 1: **Comparison of robotic manipulation paradigms.** (A) **Vision-Action (VA) Models** execute low-level actions based on visual observations. (B) **Vision-Language-Action (VLA) Models** can condition actions on language, but direct language-to-action mapping may struggle with strict compositional reasoning, long-horizon sequencing, and recovery under limited task data. (C) **Our Proposed ACE Framework** decouples reasoning from physical control. An agentic engine first decomposes complex instructions into explicit semantic sub-goals, which are then reliably executed by a downstream VA policy. This decoupled approach supports explicit long-horizon reasoning and planning while reusing a task-agnostic low-level controller.

must move beyond physical control and engage in **embodied reasoning**: inferring latent structure from language, decomposing tasks, and adapting when unexpected outcomes occur.

To achieve this task-level generalization, we present **ACE** (**A**gentic **C**ontrol for **E**mbodied manipulation), a framework for zero-shot tabletop pick-and-place. Our central thesis is that open-ended manipulation should not be treated as a new policy-learning problem for every task, but rather as **workflow reasoning**. Given a natural-language instruction, ACE uses an embodied reasoning agent equipped with two robot-facing **executable skills** for mask-mediated visual grounding and reusable pick-and-place execution. By dynamically invoking these modular skills, the agent infers and orchestrates an explicit sequence of manipulation sub-goals. Crucially, ACE maintains this workflow as an editable state rather than a rigid plan. As the task unfolds, the active sub-goal can be grounded, verified, repaired, or revised based on user feedback and execution outcomes.

To bridge the gap between high-level reasoning and low-level control, ACE introduces a **mask-mediated vision-action interface**. Rather than mapping language directly to robot actions, the agent invokes grounding skills to produce intent-level pick-and-place masks. These masks serve as stable, semantic visual priors that condition a downstream, reusable vision-action model, converting high-level agent intent into actionable visual targets.

Crucially, open-ended manipulation is prone to physical and semantic failures—grasps miss, objects shift, or user goals change. ACE addresses this through a **post-execution verification and recovery mechanism**. Rather than assuming an issued action is successful, execution outcomes become first-class inputs to the reasoning process. By checking the resulting scene state against the intended sub-goal, ACE closes the loop, automatically triggering specific skills to execute low-level retries, mask repairs, or high-level replanning. To support this continuous cycle of reasoning, perception, and recovery, ACE implements a **multi-timescale scene memory**. This architecture coordinates asynchronous modules by separating live execution state from persistent semantic-visual object identities, ensuring the agent maintains coherence when invoking skills throughout complex, multi-step tasks. By separating task-specific reasoning from reusable low-level execution, ACE handles logically complex, long-horizon tasks without task-specific retraining. We emphasize that

ACE targets task-level zero-shot generalization rather than zero-shot motor control. The downstream vision-action policy is trained only on generic mask-conditioned pick-and-place primitives and is never exposed to complete semantic tasks such as formula assembly, numerical reasoning, or constraint-based retrieval. All task-specific structure is inferred at test time by the agentic planner and composed through the reusable low-level policy. To demonstrate this zero-shot capability, we highlight the *Semantic Formula Assembly* task. Crucially, the downstream vision-action policy (π_{VA}) is only trained on generic pick-and-place primitives and has zero knowledge of mathematics, symbols, or sequential alignment. must succeed zero-shot at the task level by relying on the agent to: **1)** deduce the correct mathematical operands from an open-ended prompt (e.g., “make an equation that equals 7”), **2)** ground previously unseen symbol configurations using open-vocabulary visual grounding, and **3)** dynamically generate relative placement masks to form a coherent spatial sequence, all without any task-specific demonstrations or hardcoded waypoints.

Our core contributions are as follows:

- We formulate **open-ended tabletop pick-and-place as zero-shot, closed-loop workflow reasoning**, where an embodied agent uses internal planning and two external **robot-facing skills** to generate, ground, execute, and revise manipulation sub-goals online without task-specific training.
- We introduce a **mask-mediated vision-action interface** that bridges semantic reasoning and low-level control, allowing the system to reuse a single vision-action model across diverse logical constraints.
- We develop a **multi-timescale memory** to leverage post-execution verification and recovery mechanisms, actively using execution outcomes to trigger retries, repairs, or replanning.
- We demonstrate task-level **zero-shot generalization** on logically complex, compositional reasoning tasks, showing that shifting adaptation from low-level control to closed-loop workflow reasoning improves data-efficient generalization.

2 Related Work

2.1 Embodied Agents and Language-Guided Manipulation

Recent advancements in large language models (LLMs) and vision-language models (VLMs) have significantly advanced robotic task understanding and embodied planning [2, 6, 3, 7]. These systems excel at parsing open-ended instructions and decomposing complex tasks through iterative reasoning and tool use [8, 9, 10, 11]. However, many existing methods either stop at generating high-level textual plans, or rely on action mapping that lacks explicit memory structures and strict visual decoupling, limiting both interpretability and dynamic adaptability. While prior works emphasize agentic decomposition, ACE distinguishes itself by making closed-loop workflow replanning a first-class capability. We argue that explicit agentic planning—when coupled with a verifiable intermediate representation—is essential for zero-shot task generalization across distinct manipulation settings.

2.2 Visual Grounding and Human-in-the-Loop Control

To bridge the gap between high-level reasoning and physical execution, grounding language into visual observations is critical [12, 13]. Yet, in dynamic manipulation scenes, static grounding is insufficient; targets must be persistently tracked as the environment evolves [14]. Furthermore, ensuring safety and transparency in these complex workflows often requires human supervision and shared autonomy [15, 16, 17]. ACE unifies these domains by introducing a mask-mediated interface. Instead of opaque action mapping, our system exposes its manipulation intent through robot-usable and human-verifiable pick-and-place masks. This allows users to approve or correct sub-goals before the downstream controller is activated, combining persistent visual tracking with feedback-driven replanning to reduce target drift and execution failures.

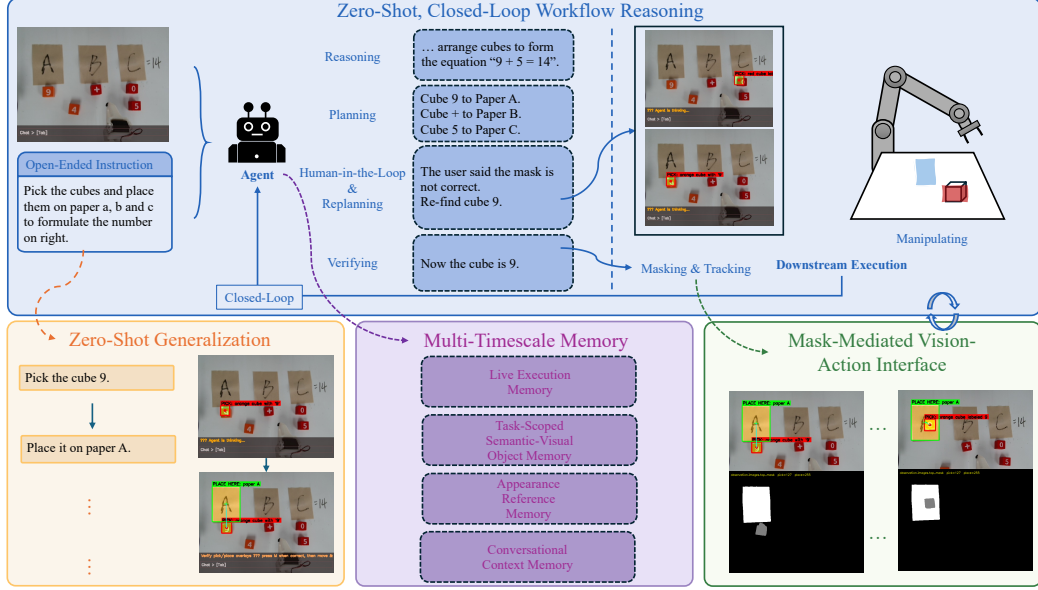


Figure 2: **Overview of the ACE framework.** The system translates open-ended instructions into physical actions through a zero-shot, closed-loop architecture. **(1) Agentic Reasoning & Planning:** The agent decomposes complex instructions into explicit semantic sub-goals. **(2) Mask-Mediated Interface & Human-in-the-Loop:** Subgoals are grounded into visual masks. Users can verify these masks before execution; if incorrect, conversational feedback triggers immediate replanning. **(3) Downstream Execution:** Approved masks are continuously tracked to guide the task-agnostic vision-action policy for physical manipulation. **(4) Multi-Timescale Memory:** The entire closed-loop process is supported by a structured memory system, enabling the agent to maintain coherence, verify outcomes, and dynamically replan.

2.3 Memory Architectures for Open-Ended Interaction

Sustaining a continuous cycle of reasoning, grounding, and correction requires robust memory architectures. Memory mechanisms—ranging from scene and object tracking to dialogue history—have become foundational for embodied systems [18, 19, 20]. In open-ended manipulation, an agent must preserve both the semantic intent of the user and the perceptual identity of the objects. To achieve this, ACE organizes these functions into a tailored multi-timescale memory hierarchy.

3 Method

We consider an interactive tabletop manipulation setting with a live image stream $I_1, I_2, \dots$ and a natural-language instruction u . ACE is built around the idea that **embodied reasoning** is the main driver of zero-shot task generalization. In this work, zero-shot refers to generalization at the task level. Let $\mathcal{D}_{\text{prim}}$ denote generic mask-conditioned pick-and-place demonstrations used to train the downstream Vision-Action (VA) policy, and let $\mathcal{T}_{\text{eval}}$ denote the semantic evaluation tasks (e.g., formula assembly). ACE assumes access to $\mathcal{D}_{\text{prim}}$ but no task-level demonstrations from $\mathcal{T}_{\text{eval}}$.

Instead of learning task-specific low-level manipulation programs, ACE formulates open-ended tabletop manipulation as a **zero-shot, closed-loop workflow reasoning** process. The system dynamically orchestrates two external robot-facing skills from language and scene context at runtime, grounding each step into human-verifiable visual targets for task-agnostic robot execution.

3.1 Zero-Shot, Closed-Loop Workflow Reasoning

The core intelligence of ACE lies in its ability to plan and adapt without task-specific retraining. Given instruction u and context c , the agentic planner generates a zero-shot workflow: $W = \mathcal{P}(u, c) = (w_1, w_2, \dots, w_K)$, where each step w_k captures an explicit semantic sub-goal rather than a raw motor command, shifting the adaptation burden away from low-level control.

Furthermore, open-ended interaction requires the workflow to dynamically adapt. ACE supports online replanning: $W' = \mathcal{R}(W, \Delta)$, where Δ includes newly observed states, user feedback, or post-execution outcomes. Replanning can be *local* (modifying the active sub-goal’s grounding) or *global* (regenerating the remaining workflow). This closed-loop adaptability ensures ACE actively revises its strategy online. **Executable Skills.** ACE exposes only two external robot-facing skills to the agent: `MaskInterface` and `PickPlace`. Other operations, including workflow decomposition, memory querying, sub-goal revision, and replanning, are handled by the agent framework as internal reasoning and state-management operations rather than as separately implemented robot skills. This design keeps the executable skill library minimal: the agent is responsible for deciding what should be manipulated and where it should be placed, while the external skills convert this intent into verified visual targets and physical robot execution.

3.2 Mask-Mediated Vision-Action Interface

The first external skill, `MaskInterface`, implements the mask-mediated vision-action interface. To bridge high-level semantic reasoning and low-level robot control, ACE employs a mask-mediated interface. For each active workflow step w_k , the system predicts an executable visual target: $M_k = \mathcal{G}(w_k, I_t, m_t)$, where I_t is the current image, m_t denotes the relevant memory state, and $M_k \in \{0, 127, 255\}^{H \times W}$ is a unified 8-bit grayscale mask. Each pixel value strictly encodes its operational role:

$$M_k(u, v) = \begin{cases} 127, & \text{if pixel } (u, v) \text{ belongs to the pick target,} \\ 255, & \text{if pixel } (u, v) \text{ belongs to the place target,} \\ 0, & \text{otherwise.} \end{cases}$$

This interface unifies three critical execution phases: **1) Human Verification:** Before action is taken, the mask M_k is rendered to the user. This iterative cycle improves safety and creates a clean division of labor: the agent handles reasoning, while the user confirms the intended spatial target. **2) Persistent Tracking:** Once approved, ACE initializes a tracker $T_k = \mathcal{T}_{\text{init}}(I_t, M_k)$ and updates it on subsequent frames to output $(T_k, M_{k,t+1}) = \mathcal{T}_{\text{update}}(T_k, I_{t+1})$. **3) Task-Agnostic Execution:** By abstracting the semantic goal w_k into a purely spatial representation, the system feeds this tracked mask directly into the downstream VA policy.

3.3 Reusable Pick-and-Place Skill and Action Chunk Generation

The second external skill, `PickPlace`, implements reusable physical execution. Given an active semantic sub-goal w_k , the skill first invokes the mask-mediated interface to ground the symbolic intent into a unified pick-and-place mask: $M_k = \mathcal{G}(w_k, I_t, m_t)$, where I_t is the current observation and m_t is the relevant memory state. After optional human verification, the mask is tracked over time to produce $M_{k,t}$ and passed to the downstream task-agnostic VA policy: $\hat{a}_{t:t+K-1} = \pi_{\text{VA}}(I_t, q_t, M_{k,t})$, where K is the action-chunk horizon and q_t is the robot proprioceptive state. At inference time, ACE executes predicted actions in a receding-horizon manner while continuously updating the visual observation and tracked mask. This skill is reusable because it contains no task-specific semantics. It does not reason about arithmetic, numerical constraints, object roles, or long-horizon task structure. The agentic planner determines the semantic sub-goal, while the mask interface converts that intent into robot-usable visual targets. After execution, the skill returns an execution report to the workflow manager, enabling ACE to advance, retry, repair the mask, or replan.

3.4 Multi-Timescale Memory for Robust Recovery

ACE operates as an asynchronous system where planning, tracking, rendering, and user interaction proceed on different timescales. To coordinate these processes and support post-execution verification, ACE relies on a shared scene-state manager backed by a multi-timescale memory architecture: **Live execution memory:** Maintains the current workflow position, active masks, tracker state, and execution progress on fast timescales. **Task-scoped semantic-visual memory:** Maintains object names, aliases, roles, and visual associations, supporting identity consistency across user corrections. **Appearance reference memory:** Stores image features or crops, enabling identity-preserving reacquisition if a target leaves the field of view. **Conversational context memory:** Maintains a bounded dialogue history, allowing the planner to interpret user feedback within the current task setting. By centralizing state management through this memory hierarchy, ACE preserves semantic and perceptual consistency, effectively closing the loop between planning, execution, and recovery. Leveraging this memory architecture, the agent verifies the successful execution of each step by computing the spatial overlap between the target masks and cross-referencing it with the updated visual observation. Once the agent finds the error, it will replan the workflow to follow the instruction.

4 Experiments

4.1 Experiment Setup Details

To systematically evaluate the capabilities of the ACE framework, we design a comprehensive experimental suite that tests both high-level cognitive reasoning and low-level physical execution. We deploy the agentic reasoning engine on a cloud server equipped with 4 NVIDIA RTX 3090 GPUs, while the mask-mediated interface and downstream vision-action (VA) policy run locally on a single RTX 3090 to ensure low-latency physical control. To improve reliability, all quantitative results for both ACE and the baselines are averaged over 10 independent trials for each task. Furthermore, in every trial, we randomize both the initial spatial configuration of the objects and the phrasing of the instructions, while ensuring that the underlying semantic meaning remains identical. The ACE implementation details are in Appendix B.

4.2 Tasks

We evaluate ACE on logically complex tabletop tasks requiring open-ended reasoning and execution. These tasks differ in semantic constraints, object selection logic, and workflow structure, providing a rigorous test of zero-shot task generalization: **Semantic formula assembly:** The system must perform multi-step sequential manipulation, zero-shot reasoning over specific number and operator cubes to physically construct a requested mathematical equation. **Constraint-based retrieval:** The system must evaluate logical or numerical constraints to identify specific target objects and transport them to a designated spatial region. Examples are shown in Appendix A.

4.3 Baselines

Our goal is to evaluate task-level zero-shot generalization rather than zero-shot motor skill learning. To provide a conservative comparison, ACT [21] and $\pi_{0.5}$ [22] are trained or fine-tuned using approximately one hour of full-task demonstrations with Lerobot [23], including complete formula assembly and constraint-based retrieval trajectories. In contrast, ACE’s downstream VA policy is trained for approximately one hour only on generic mask-conditioned pick-and-place primitives and is never exposed to complete semantic task trajectories. Thus, the baselines receive stronger task-specific supervision, while ACE must compose reusable primitives through explicit workflow reasoning at test time.

5 Results

We present the quantitative evaluation of ACE against the baselines in Table 1. The results suggest that, under our low-data task-supervised setting, direct policy-learning baselines struggle to acquire the explicit compositional reasoning required by these tasks, yielding a 0.0% Success Rate (SR) across the board. Specifically, while the standard Vision-Action (VA) baseline (ACT) manages to learn basic motor behaviors (i.e., it knows *how* to grasp), it does not explicitly model the semantic task structure required for selecting the correct operands, operators, or constraint-satisfying cubes. Its marginal partial scores (an average FGS of 3.2 out of 20) mainly arise from occasional successful grasps or placements that do not consistently correspond to the intended semantic target. On the other hand, the Vision-Language-Action (VLA) baseline ($\pi_{0.5}$) similarly fails to achieve end-to-end task success (0.0% SR, 3.0 FGS) in our setup. Because these direct policies do not expose intermediate symbolic or visual sub-goals, diagnosing whether a failure comes from semantic selection, grounding, or control is difficult. In stark contrast, ACE exposes an interpretable workflow through explicit sub-goals and human-verifiable masks, achieving a 50.0% SR in Semantic Formula Assembly (with a high FGS of 17.8) and a 70.0% SR in Constraint Retrieval. By decoupling cognition from control, our agent reliably performs zero-shot reasoning to determine precise pick-and-place subgoals, maintaining a robust 90.0% Grounding Accuracy (GA) across both tasks. These semantic decisions are then seamlessly grounded and executed through our mask-mediated vision-action interface, allowing the system to robustly complete long-horizon tasks. Importantly, this transparent architecture allows us to precisely diagnose our system’s failure modes. A detailed analysis of these failure modes is provided in the Limitations section (Section 7).

Table 1: **Quantitative Results on logically complex tabletop tasks.** We report the overall Success Rate (SR), the Fine-Grained Score (FGS, max 20) (Appendix C), and Grounding Accuracy (GA), which indicates the grounding accuracy of the pick and place masks for the Formula Assembly task. For Constraint Retrieval, we report SR and GA. Best results are highlighted in **bold**. Each result is averaged over 10 physical trials per task. Note that GA specifically measures the vision-language model’s ability to successfully localize and mask the agent’s intended target, independent of whether the high-level semantic reasoning was correct.

Method	Semantic Formula Assembly			Constraint Retrieval	
	avg. FGS (Max 20)	SR (%)	GA (%)	SR (%)	GA (%)
ACT (Imitation)	3.2	0.0	–	0.0	–
$\pi_{0.5}$ (End-to-End VLA)	3.0	0.0	–	0.0	–
ACE	17.8	50.0	90.0	70.0	90.0

6 Ablations

To validate the individual components of our framework, we analyze several ablative settings (detailed in Appendix D) based on the quantitative results in Table 2.

First, at the policy level, we isolate our evaluation to the **grasp success rate** to observe the critical role of our mask-mediated interface in visual generalization. When raw RGB images are fed directly into the downstream Diffusion Policy alongside the masks (*Mask + Original Image*), grasping performance plummets to a mere 30.0% in the presence of unseen object colors or novel backgrounds. This severe degradation indicates that the policy overfits to the raw RGB input, making it highly vulnerable to domain shifts. In contrast, relying strictly on our mask-centric representation (*Mask Only*) effectively filters out irrelevant environmental noise and spurious visual features, achieving a highly robust 90.0% grasp success rate. Second, at the system level, removing the agentic reasoning engine (*w/o. Agentic Planner*) reduces the framework to a standalone Diffusion Policy (DP) trained on the exact same demonstration data as the ACT baseline. In this setting, end-to-end task performance drastically degrades to a 0.0% Success Rate across both tasks, with a marginal average FGS of 3.4. This complete failure underscores the absolute necessity of explicit high-level planning for

Table 2: **Ablation Study.** We evaluate the framework across two levels. **Top:** Policy-level ablation comparing the downstream Diffusion Policy (DP) grasp success rate when conditioned on the mask alone versus the mask plus the original RGB image. **Bottom:** System-level ablations evaluating the impact of the reasoning agent and human-in-the-loop verification on end-to-end task performance. Best results are highlighted in **bold**.

(a) Policy-Level Ablation (Visual Input)					
Ablation Setting			Grasp Success Rate (%)		
DP (Mask + Original Image)			30.0		
DP (Mask Only) [Ours]			90.0		
(b) System-Level Ablation (Reasoning & Verification)					
Ablation Setting	Semantic Formula Assembly			Constraint Retrieval	
	avg. FGS (Max 20)	SR (%)	GA (%)	SR (%)	GA (%)
ACE (w/o. Agentic Planner)	3.4	0.0	–	0.0	–
ACE (w/o. Human Verification)	13.2	30.0	70.0	20.0	60.0
ACE (Full Framework)	17.8	50.0	90.0	70.0	90.0

logically complex, long-horizon tasks. Finally, returning to the system level, we evaluate ACE in a fully autonomous mode (*w/o Human Verification*). While the agent remains highly capable, the absence of human oversight leads to a measurable drop in overall task success—falling from 50.0% to 30.0% in Formula Assembly, and from 70.0% to 20.0% in Constraint Retrieval. This decline is primarily attributed to occasional inaccuracies in the agent’s spatial placement predictions, which is reflected by the Grounding Accuracy (GA) dropping from 90.0% down to 70.0% and 60.0% for the respective tasks. This finding strongly reinforces the practical value of our verifiable interface, which allows users to intuitively catch and correct minor grounding errors before physical execution, thereby maximizing system reliability.

7 Limitations and Future Work

ACE has several limitations. First, the current system depends on human approval before robot execution, which improves safety but introduces interaction overhead. Second, the quality of the workflow depends on the reasoning ability of the planner and the fidelity of visual grounding. Third, the present formulation is centered on pick-and-place and may require further extension for richer contact-rich manipulation. Lastly, our baseline comparison focuses on a low-data regime in which all learned controllers receive approximately one hour of demonstrations. Scaling task-level demonstrations for VLA baselines may improve their performance. Nevertheless, our results highlight the data efficiency of explicit workflow reasoning when only primitive-level robot data are available. Our analysis reveals that the vast majority of ACE’s failures stem from perceptual grounding or physical execution rather than high-level cognitive planning. Specifically, errors predominantly occur when the agent generates imprecise visual masks (grounding failures) or when the downstream Diffusion Policy (DP) struggles to execute a stable grasp (low-level control failures). Notably, in the more complex Semantic Formula Assembly task, we observe a slight increase in reasoning errors due to the heightened logical difficulty, which accounts for its lower overall Success Rate (SR) compared to Constraint Retrieval. Nevertheless, logical reasoning errors remain infrequent across all trials. This error distribution validates the robust zero-shot cognitive capabilities of our agentic engine and underscores the practical advantages of a decoupled, interpretable architecture. Future work should explore reducing approval burden through calibrated confidence estimation, extending workflow reasoning to broader manipulation skills, and studying stronger memory mechanisms for long-horizon embodied interaction.

8 Conclusion

In this work, we present ACE, an interpretable framework that achieves zero-shot generalization by decoupling high-level cognitive planning from low-level physical control. ACE translates semantic subgoals into a mask-mediated interface, providing stable spatial priors for a task-agnostic policy and enabling human-in-the-loop verification. Supported by a multi-timescale memory, the system dynamically tracks objects and autonomously recovers from failures. Evaluations on logically complex tasks demonstrate that ACE’s explicit reasoning and verifiable representations significantly outperform end-to-end baselines, offering a practical step toward open-ended, adaptable embodied manipulation.

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A Task descriptions

Figure 3 shows demonstrations of our tasks.

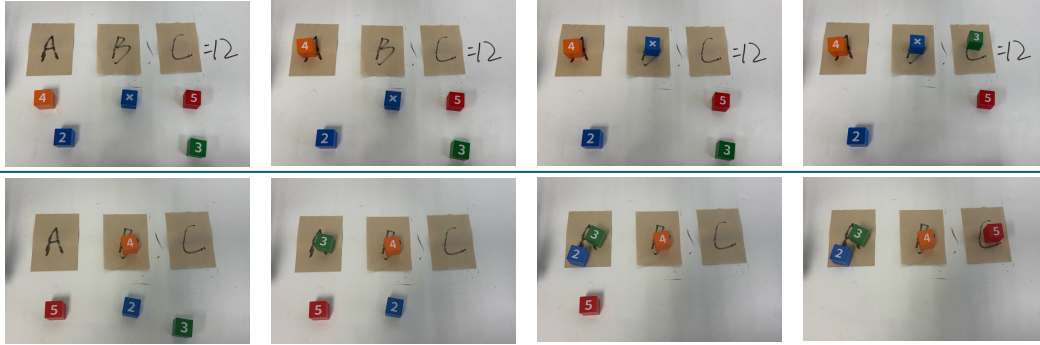


Figure 3: **Execution sequences of open-ended manipulation tasks.** From left to right, the frames capture the progression of ACE completing multi-step objectives. **Top:** Semantic Formula Assembly, requiring sequential spatial arrangement of mathematical components. **Bottom:** Constraint-Based Retrieval, where the agent visually filters and sorts cubes (e.g., values < 4 to the left, > 4 to the right) based on logical conditions. In our ACE experiment, we can

B Implementation Details

While the ACE framework is model-agnostic, our physical instantiation relies on several state-of-the-art foundation models to realize the agentic workflow and visual grounding.

Agentic Planner and Replanner: We implement the core language agent using the Qwen-Agent framework [24], powered by Qwen3.6-35B-A3B [25]. Qwen processes both the user instructions and the rendered scene context to generate and revise the workflow steps. Crucially, the executable skills described in our framework (e.g., visual grounding, state querying, and robot execution) are registered as callable tools within Qwen-Agent, allowing the model to actively orchestrate the physical workspace.

Mask-Mediated Vision-Action Interface: For the mask-mediated grounding ($\mathcal{G}$) and persistent tracking ($\mathcal{T}$), we utilize SAM3 [26] to generate high-fidelity pick-and-place masks and use Cutie [27] to track the masks consistently.

Robot Control: The human-verified masks are dispatched to a SO-101 arm [28]. The low-level execution is handled by Lerobot [23] with Diffusion Policy (DP) [29]. We train the DP controller

for approximately one hour using only generic mask-conditioned pick-and-place demonstrations. These demonstrations contain random object-to-region transfers specified by pick/place masks and do not include formula assembly, numerical comparison, constraint-based retrieval, or complete multi-step semantic task trajectories. Therefore, the downstream controller provides only a reusable manipulation primitive; all task-level reasoning and sequencing are produced by ACE at inference.

C Fine-grained Scoring

To comprehensively evaluate both high-level reasoning and low-level execution, we track standard metrics including step grounding accuracy, approval efficiency, and replanning rate. However, for long-horizon tasks like **semantic formula assembly**, binary success rates are insufficient to capture execution nuances. Therefore, we introduce a fine-grained step-wise evaluation protocol.

Each episode consists of three sequential pick-and-place steps (first operand, operator, second operand). Each step is assigned a maximum score of 6 points:

- **2 points** for selecting the correct cube (Semantic reasoning) or the manipulator move upon it and try to grasp it;
- **2 points** for successfully grasping the selected cube (Physical execution);
- **2 points** for accurately placing the block into the target slot (Spatial grounding).

We additionally assign 2 points for final equation correctness, which evaluates whether the assembled expression mathematically matches the target value. The total score for one episode is computed as:

$$S_{max} = 3 \times (2 + 2 + 2) + 2 = 20.$$

D Ablation Setup Details

To isolate the contributions of our specific architectural choices, we evaluate the framework under three ablative settings:

- **DP (Mask + Original Image):** To evaluate the robustness of our low-level control, we modify the vision-action interface by feeding the raw RGB image alongside the mask into the downstream DP. This policy-level ablation specifically measures the **grasp success rate** to test the necessity of our purely spatial, mask-only representation for visual generalization. We conducted 10 physical trials to measure the success rate, using the pick-and-place masks provided by ACE.
- **ACE (w/o. Agent):** We disable the high-level reasoning engine, reducing the system to a standalone Diffusion Policy (DP). This tests whether the downstream policy can implicitly learn long-horizon task semantics without explicit workflow decomposition.
- **ACE (w/o. Human):** We bypass the human-in-the-loop verification step, triggering physical execution immediately after the agent grounds the subgoal. This measures the agent’s raw hallucination rate and quantifies the safety margin provided by human oversight.